AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

M218
Relational databases

Final Examination
Fall Semester 2024/2025

Date: 11 /Jan./2025

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mber of Exam Pages: 3 luding this cover sheet)	**Time Allowed:** 2 hours

structions:

- Total Marks:50
- This exam consists of 2 parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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Part I [10 Marks]

This part includes 5 MCQ; each weights 2 marks. Choose the most suitable answer for the following questions:

1- The SQL clause used to filter grouped records is:
   a. WHERE
   b. FILTER
   c. HAVING
   d. RESTRICT

2- To create a view in SQL, we use:
   a. CREATE VIEW
   b. MAKE VIEW
   c. BUILD VIEW
   d. SET VIEW

3- The SQL command to remove privileges is:
   a. DELETE PRIVILEGE
   b. REMOVE
   c. REVOKE
   d. SUBTRACT

4- What does the GRANT OPTION in SQL allow a user to do?
   a. Update data
   b. Grant privileges to other accounts
   c. Create new tables
   d. Delete records

5- Which principle in the DBA Code of Ethics involves judging appropriate actions for a given time?
   a. Prudence
   b. Justice
   c. Temperance
   d. Courage

Part II [40 Marks]

Question 1 [25 Marks]

The "Metro Ticketing" relational database is described by the following relation schemas:

PASSENGERS (PASSENGERID, PASSENGERNAME, CATEGORY, CITY)
STATIONS (STATIONID, STATIONNAME, LINE, LOCATION)
TICKETS (TICKETID, PASSENGERID, TICKETDATE)
TICKETDETAIL (TICKETID, STATIONID, TICKETQTY)

You are asked to write the following queries in SQL:

1. List the names of passengers (PASSENGERNAME) who have purchased at least one ticket in 2024. [2 marks]
2. List the station names (STATIONNAME) appearing in the ticket number 560789 with a quantity greater than 3. [2 marks]
3. Write an SQL query to find all stations whose names start with 'Central'. [3 marks]
4. Provide the list of station names (STATIONNAME) where tickets were purchased in September 2023 but not in August 2023. [3 marks]

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5. Provide the list of station names (STATIONNAME) visited by passengers from "Beirut". [3 marks]
6. Provide the list of passenger IDs (PASSENGERID) who did not purchase any tickets in September 2023. [3 marks]
7. Provide the list of passenger IDs (PASSENGERID) of passengers from "Tripoli" who have purchased more than 15 tickets in total in 2023. [3 marks]
8. List all unique cities where passengers have purchased tickets. [3 marks]
9. Write an SQL query to list all distinct passenger IDs and station IDs together from the PASSENGERS and STATIONS tables. [3 marks]

Question 2 [15 Marks]

The "University Management" relational database is described by the following relation schemas:

STUDENTS (StudentID, StudentName, Major, City)
COURSES (CourseID, CourseName, Department)
ENROLLMENTS (EnrollmentID, StudentID, CourseID, EnrollmentDate)
FACULTY (FacultyID, FacultyName, Department)
GRADES (GradeID, StudentID, CourseID, Grade)

Use the appropriate SQL DCL script to perform the following actions:

a) Allow the user T1 to insert and update records in the COURSES table. Additionally, enable T1 to grant these privileges to other users. **[2 marks]**
b) Remove the UPDATE privilege on the COURSES table from user T2. **[2 marks]**
c) Grant the user T3 the ability to select only StudentID, StudentName, and Major from the STUDENTS table for all students majoring in 'Computer Science'. **[3 marks]**
d) The Research Department needs to analyze student performance based on data from the GRADES table but should only access data related to students in the 'Mathematics' major. **[4 marks]**
    i. Create a view named MathematicsPerformance that includes StudentID, StudentName, Grade, and Major.
    ii. Ensure that user T4 can query this view but cannot grant access to this view to others.
e) You are tasked with providing restricted access to the ENROLLMENTS table, where only records of enrollments made in the year 2022 should be visible to user T5. Additionally, T5 should only be able to see the EnrollmentID, CourseID, and EnrollmentDate columns. **[4 marks]**
    i. Formulate the SQL statement to create a view that meets these requirements.
    ii. Grant the necessary permissions to T5.

End of questions

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